

Master Thesis Presentation

- Good afternoon, my name is Heykel Khadhraoui. I developed a surface grinding controller for a seven-joint robot.
- Normal pressing and compliant rotation allow contact to move the tool towards alignment. A separate experiment tests how the arm responds to disturbances.

Motivation

- Grinding needs the tool face aligned with the physical surface. The configured and physical surface orientations have an unknown difference.
- With a tilted tool, normal pressing creates a contact moment. Low rotational stiffness lets this moment turn the tool towards alignment.
- Cartesian impedance makes position and rotation compliant. The desired orientation stays fixed while the rotational spring and damper remain active.

Overview

- I will explain the controller, introduce the experiments, and present the validation and results. I will finish with conclusions and future work.

Cartesian pose

- The end effector (EE) is the tool or gripper at the end of the arm. Its Cartesian pose describes position and rotation relative to the robot base frame.
- Position uses x, y, and z, while roll, pitch, and yaw illustrate rotation. Together they give six degrees of freedom.
- Forward kinematics calculates the pose from all seven joint angles, with $p = p(q)$ for position. The controller uses a rotation matrix for orientation.

Cartesian impedance controller

- Stiffness acts like a spring: for the same error, higher stiffness gives a larger restoring force or moment.
- Position errors generate forces, and orientation errors generate moments. Damping opposes differences from the desired linear and angular velocities.
- Force and moment form a Cartesian wrench, controlled by six-by-six matrices with three-by-three blocks. The zero off-diagonal blocks mean translation and rotation are decoupled here.

Real-time control

- The robot receives joint torques. We generate the Cartesian contribution from the wrench using the Jacobian transpose:

$$\tau_{\text{cart}} = J^{\top}(q)F$$

- The Jacobian maps joint velocities to the EE's linear and angular velocities. Model compensation and enabled null-space torques complete the command.
- The measured pose feeds back into the control loop. The C++ controller uses libfranka and updates every millisecond.

Translation–rotation coupling

- The tool centre point (TCP) is the controlled reference point attached to the EE. The centre of compliance (CoC) at p_c is a virtual reference for stiffness and damping.
- Moving the CoC changes how position and rotation errors generate force and moment. Its displacement from the TCP is r_c :

$$r_c = p_c - p_{\text{TCP}}$$

- For small rotations, we express the shifted stiffness and damping at the TCP. Non-zero off-diagonal blocks show the coupling:

$$K_{\text{TCP}} = A^\top K_c A,$$

$$D_{\text{TCP}} = A^\top D_c A.$$

Centre of compliance (CoC)

- Shifting the CoC can change both the commanded force and moment. The shifted controller's force acts as if applied at p_c :

$$m = m_R + r_c \times f$$

- The added moment can support or oppose alignment. For the same force, reversing the displacement reverses this contribution.
- With the CoC at the TCP, the added contribution vanishes. The rotational spring and damper still produce m_R .

Null-space controller

- Alongside the tool response, we control the arm's extra motion. At full Jacobian rank, seven joints leave one null-space direction with zero instantaneous EE motion:

$$J(q)\dot{q}_{\text{null}} = 0$$

- Damping opposes this redundant joint motion.
- A second term seeks easier motion in the weakest EE direction. This objective is called conditioning and can initiate arm motion.
- At a singularity, the Jacobian loses rank and the EE loses an instantaneous motion direction. Near it, some EE motions require very large joint velocities.

Contact experiment

- The process goes from tool orientation and surface approach to contact establishment, then grinding. These experiments evaluate the five-second contact-establishment phase.

- Low normal stiffness allows compliant pressing, while higher tangential stiffness resists sideways movement. Rotational stiffness is higher about the normal and lower about the surface tangents.
- The position reference advances into the surface to generate pressing force. The desired orientation stays at its entry value, so contact moments drive the rotation.

Angular quantities

- The measured angular offset compares the entry orientation with the configured surface. It uses measured EE orientation and the calibrated tool normal.
- The contact response is defined from the measured end orientation back to entry. Its sign is therefore opposite to the EE's actual start-to-end rotation.
- Positive rotation is anticlockwise in the sketch, and an aligning response has the same sign as the offset. These quantities do not directly measure the physical tool–surface angle.

Null-space experiment

- The desired position and orientation stay fixed while joint torques apply a repeatable disturbance. This is equivalent to a virtual 20-newton point force on link three.
- We compare no null-space torque, damping, and two conditioning settings, with three trials each. Conditioning is already active during the five seconds before the disturbance.
- All results use seconds five to nine, displayed as zero to four after disturbance onset. We compare total projected joint motion, minimum singular value, and net projected displacement.

Normal-force plausibility assessment

- I manually push the tool about 20 millimetres along the surface normal, with the CoC at the TCP. This checks the spring-force prediction with the desired pose fixed.
- The stiffness is 1000 newtons per metre and the measured position-error change is -19.650 millimetres. Their product gives the spring-force prediction:

$$F_{n,qs} = (1000 \text{ N/m}) \cdot (-0.019650 \text{ m}) = -19.650 \text{ N}$$

- In the shaded stationary interval, the commanded mean is -19.640 newtons, only 0.05 percent from the calculation. The robot's model estimate is -22.292 newtons, 13.50 percent from the command.

Moment plausibility assessment

- I manually rotate the tool about surface tangent one while keeping the normal push. This checks the spring-moment prediction with the desired pose fixed and CoC at the TCP.
- The measured rotational error changes by 6.565° relative to the loaded baseline. With a stiffness of 15 newton metres per radian, converting this change to radians gives:

$$M_{t1,qs} = (15 \text{ N m/rad}) \cdot (6.565 \cdot \frac{\pi}{180} \text{ rad}) \approx 1.719 \text{ N m}$$

- In the shaded stationary interval, the commanded mean matches the calculated 1.719 newton metres. The robot's model estimate is 1.852 newton metres, 7.75 percent from the command.

Commanded and estimated wrench

- A separate contact-establishment trial compares commanded force and moment with the robot's model estimates. The CoC is at the TCP.
- During the final stationary interval, mean differences are 2.32 percent for force and 4.87 percent for moment. The curves differ more during motion.

Baseline contact response

- With the CoC at the TCP, both larger entry offsets produced EE rotation towards the configured surface. The nominal-zero condition showed only a small response.
- Entry offsets of $+9.31^\circ$ and -9.41° give responses of $+7.57^\circ$ and -10.83° .
- The contact comparisons show three-trial means. Physical alignment is assessed indirectly from EE rotation.

Effect of stiffness

- Higher rotational stiffness opposes contact-induced rotation. Increasing it from 5 to 50 newton metres per radian reduced the response by about 64 percent.
- Across the tested tangential stiffness range, the response varied by only 0.03° . Only one tangent was varied, while the other stayed fixed.

Effect of CoC position

- For a positive entry offset, plus 40 millimetres supports contact rotation and minus 40 millimetres nearly suppresses it.
- For a negative entry offset, the supporting displacement reverses.
- In the tested conditions, supporting displacement increases the response by about 4 percent. Opposing displacement reduces it by about 98 to 99 percent.

Contact response and interaction wrench

- In these traces, contact rotation settles in about 2.3 seconds with supporting CoC displacement, compared with 3.6 seconds at the TCP.
- Final estimated normal forces are similar, at about -83 , -79 , and -78 newtons.
- The interaction moment reverses with CoC displacement. These forces and moments come from the robot's model estimate.

Cumulative null-space motion

- Cumulative motion, E_N , adds projected joint movement in both directions. These degrees describe arm motion, not rotation of the tool.

- Damping reduces it from 7.598° to 5.689° , about 25.1 percent. Conditioning gives 0.288° at $k_\sigma = 1.5$ and 1.687° at $k_\sigma = 2$.
- Curves show three-trial means over the displayed four-second interval. Shading shows one sample standard deviation.

Jacobian conditioning

- A smaller minimum singular value means some tool motions require larger joint velocities. Zero means the EE loses an instantaneous motion direction.
- It falls with no null-space torque and with damping. Both conditioning settings keep it nearly constant during the disturbance.
- Curves show three-trial means over the same four-second interval and use the same Jacobian convention. Shading shows one sample standard deviation.

Net null-space displacement

- Net displacement measures projected joint movement remaining after opposite motions cancel. It is 7.516° without null-space torque and 5.609° with damping.
- Mean conditioning displacements are only 0.015° and -0.011° . Conditioning counters the disturbance in the tested direction.
- Cumulative motions are 0.288° and 1.687° . The larger value is consistent with more back-and-forth motion despite similarly small net displacement.

Conclusion

- Contact drove EE rotation towards alignment while the desired orientation stayed fixed. Physical alignment was assessed indirectly.
- Within the tested rotational stiffness range, higher stiffness reduced contact rotation by about 64 percent.
- Tested tangential stiffness changes had little effect: only 0.03° variation. The other tangent remained fixed.
- CoC displacement supports or opposes rotation. At the TCP, its added moment vanishes while the rotational spring and damper remain active.
- Damping reduced arm motion, and conditioning countered the tested disturbance direction. Nearly zero net displacement coexisted with back-and-forth motion.
- Future work is a more rigid tool mount and adaptive CoC selection. After alignment, the adaptive CoC should return to the TCP.

Null-space torques and projector

- The kinematic projector selects null-space torque directions. It uses the Jacobian pseudoinverse J^+ .

$$N_\tau = I_7 - J^\top J^+\top$$

- Damping and conditioning add to give the total null-space torque.

$$\tau_{\text{null}} = \tau_d + \tau_\sigma$$

- Damping opposes projected joint velocity and is zero at rest. d_{null} is the damping coefficient.

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}$$

- Conditioning probes both directions along the unit null-space vector v_7 . It chooses the direction with the larger minimum singular value.

$$\tau_\sigma = k_\sigma N_\tau (s_\sigma v_7)$$

- The selector s_σ is +1 or -1, and zero within the deadband. k_σ sets the torque magnitude, which can be nonzero at rest.

Contact demonstration

- The video shows pickup, approach, contact, and the pre-grinding hold.
- During the hold, I change the surface orientation. Normal pressing continues as contact moments rotate the tool.
- The desired orientation stays fixed, with the CoC at the TCP. The rotational spring and damper remain active.

Disturbance demonstration

- The controller holds the tool pose throughout the video. Without null-space torque, the disturbance moves the arm.
- Damping opposes the joint motion.
- Conditioning actively changes the arm configuration and can initiate motion. The tool stays approximately in place.